Project Demonstration

Scalability & Future Plan

Date: 10 July 2026
Team ID: SWTID-2026-6119
Project Name: Credit Card Approval Prediction System
Maximum Marks: 2 Marks

Current Scalability

- Modular pipeline: preprocessing (encoders, scaler) and the model are saved separately from the application code, so the model can be retrained on a larger dataset and swapped in without changing app.py.
- Stateless prediction requests: each prediction is independent and requires no session state, so the Flask app can be scaled horizontally (multiple instances behind a load balancer) if traffic increases.
- Fast inference: average prediction time of 5.4 milliseconds means the current architecture can comfortably handle a high volume of concurrent requests before performance becomes a concern.

Future Enhancements

Enhancement	Description	Priority
Cloud Deployment	Deploy the Flask app to IBM Cloud (or AWS/Azure) with a production WSGI server (e.g. Gunicorn) instead of the local development server.	High
Explainable AI	Integrate SHAP or LIME to show applicants and officers which factors most influenced a specific decision, improving transparency and trust.	High
Larger, updated dataset	Retrain periodically on a larger, more recent dataset to keep the model's predictions aligned with current lending patterns.	Medium
Officer Dashboard	Add an admin view where bank officers can see historical predictions, override decisions, and track approval trends over time.	Medium
Authentication & Access Control	Add login for bank staff so only authorized users can access sensitive parts of the system.	Medium
API Endpoint	Expose a REST API (in addition to the web form) so the prediction engine can be integrated directly into a bank's existing core systems.	Low
Fairness Auditing	Periodically test the model for bias across demographic groups to ensure compliance with fair-lending principles.	High

Long-Term Vision

Beyond credit card approval, the same architecture (data pipeline + model comparison + Flask serving layer) could be extended to related banking decisions - personal loan approval, credit limit recommendations, or early risk flagging on existing accounts - making this project a reusable foundation rather than a single-purpose tool.